

EMIL PAUL

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PROFESSIONAL SUMMARY

Computer Science BSc candidate with professional software and data engineering experience at the **Ocean Frontier Institute**, building Python data pipelines, backend APIs, PostgreSQL services, and agentic LLM tools. Brings 3+ years of shift-lead experience in high-precision manufacturing, with strengths in troubleshooting, quality control, and technical ownership.

EXPERIENCE

Software Developer Intern

May 2026 - Aug 2026

Ocean Frontier Institute, Dalhousie University (COVE Internship)

Halifax, NS

- Built and extended an authenticated FastAPI and PostgreSQL feedback system for an LLM knowledge application, implementing submission and admin-retrieval endpoints, role-based access control, filtering, pagination, database migrations, parameterized SQL, and automated unit and integration tests.
- Implemented and merged an ORCID publication-retrieval tool for an agentic LLM knowledge assistant using Python, Pydantic, Neo4j/Cypher, and the ORCID API, with deterministic identity resolution, checksum validation, structured citations, and comprehensive automated testing.
- Developed and enhanced Python-based scientific data pipelines using pandas, xarray, ioos_qc, ERDDAP, and NetCDF to automate quality control, missing-value handling, dataset transformations, and configurable output workflows across multiple oceanographic data products.
- Created reusable QC helper tooling and climatology workflows for five QARTOD tests (gross range, spike, flat-line, rate-of-change, and climatology), including automated threshold generation and location-aware temporal validation for CODAR data.
- Contributed React/TypeScript functionality to the knowledge-assistant frontend, developing typed API contracts, authenticated client integration, feedback UI components, and responsive desktop/mobile navigation.

ORCID Repository Project Contributor

Fall 2025

Canadian Research Knowledge Network (CRKN) - Academic-Industry Collaboration

- Contributed bug fixes, repository maintenance, and system updates to an ORCID-focused data repository during a semester-long collaboration with CRKN.
- Worked within an existing codebase and external stakeholder requirements, using Git-based version control and collaborative software-maintenance practices to deliver repository updates.

Shift Lead Hand / CNC Programmer / Machinist Apprentice

Nov 2020 - Jan 2024

JessEm Tool Corporation

Moncton, NB

- Led shift operations in a high-precision manufacturing environment, balancing safety, quality, and production requirements while diagnosing machine, tooling, and process issues under time constraints.
- Programmed CNC equipment from engineering drawings and collaborated with engineers and technicians to troubleshoot production problems, improve workflows, and maintain tight manufacturing tolerances.

EDUCATION

Mount Allison University

Expected Apr 2027

BSc in Computer Science; Minors: Mathematics, Data Science, Visual & Material Culture Studies

Sackville, NB

CGPA: 3.7 | Dean's List 2025-2026

Georgian College

2018 - 2020

Mechanical Technician - Precision Skills

Barrie, ON

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C, Bash

Backend & Databases: FastAPI, PostgreSQL, Neo4j, Cypher, Pydantic, REST APIs, Node.js

AI/LLM: OpenAI API, retrieval-augmented generation (RAG), agentic LLM systems, Neo4j graph/vector retrieval, LLM evaluation

Data Engineering: pandas, NumPy, xarray, NetCDF, ERDDAP, ioos_qc, QARTOD, CF/ACDD standards

Frontend: React

Tools & Practices: Docker, Docker Compose, Git/GitHub, pytest, Flyway, Linux, test-driven development, unit and integration testing, code review